

Experiment analysis step 2: primary crossover analysis

Paired descriptives, main figures, paired contrast, and fixed-effects model

W241 Plank Study

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Contents

0.1	Purpose	1
0.2	Load analytic datasets	1
0.3	Analytic sample	2
0.4	Descriptive summaries	2
0.5	Compact metrics summary	3
0.6	Exploratory figures	4
0.7	Model 1: raw paired contrast	6
0.8	Model 2: fixed-effects crossover regression without period adjustment	7
0.9	Model 3: fixed-effects crossover regression with period adjustment	8
0.10	Main-effect comparison figure	9
0.11	Verification	10

0.1 Purpose

This document estimates the primary within-subject effect of **high_tempo** versus **low_tempo** music on plank time-to-exhaustion using the canonical datasets created in Step 1.

The result sequence follows the report plan:

1. paired descriptives
2. paired mean-difference estimate
3. participant fixed-effects model without a period indicator
4. participant fixed-effects model with a period indicator

0.2 Load analytic datasets

```
paired <- sanitize_dt_ascii(fread(derived_path("paired_analysis.csv")))
long_dt <- sanitize_dt_ascii(fread(derived_path("long_repeated_measures.csv")))
participant_flow <- sanitize_dt_ascii(fread(derived_path("participant_flow.csv")))

long_primary <- long_dt[in_primary_analysis == 1L]

stopifnot(nrow(paired) > 0L)
stopifnot(uniqueN(paired$participant_id) == nrow(paired))
stopifnot(nrow(long_primary) == 2L * nrow(paired))
stopifnot(uniqueN(long_primary$participant_id) == nrow(paired))
```

0.3 Analytic sample

```
sample_tab <- data.table(  
  quantity = c(  
    "Paired analytic participants",  
    "Repeated-measures rows used in FE model",  
    "High-tempo first",  
    "Low-tempo first"  
  ),  
  value = c(  
    nrow(paired),  
    nrow(long_primary),  
    paired[, sum(crossover_order == "high_tempo_first")],  
    paired[, sum(crossover_order == "low_tempo_first")]  
  )  
)  
  
kable(sample_tab, caption = "Primary crossover sample")
```

Table 1: Primary crossover sample

quantity	value
Paired analytic participants	38
Repeated-measures rows used in FE model	76
High-tempo first	19
Low-tempo first	19

0.4 Descriptive summaries

```
desc_tab <- data.table(  
  quantity = c(  
    "Mean high-tempo TTE",  
    "Mean low-tempo TTE",  
    "Mean paired difference (high - low)",  
    "SD paired difference",  
    "Median paired difference",  
    "IQR paired difference",  
    "Independent participants"  
  ),  
  value = c(  
    mean(paired$high_tempo_tte),  
    mean(paired$low_tempo_tte),  
    mean(paired$diff_high_minus_low),  
    sd(paired$diff_high_minus_low),  
    median(paired$diff_high_minus_low),  
    IQR(paired$diff_high_minus_low),  
    nrow(paired)  
  )  
)  
  
desc_tab[, value := round(value, 3)]  
kable(desc_tab, caption = "Paired descriptive summaries")
```

Table 2: Paired descriptive summaries

quantity	value
Mean high-tempo TTE	102.440
Mean low-tempo TTE	88.071
Mean paired difference (high - low)	14.369
SD paired difference	28.921
Median paired difference	8.708
IQR paired difference	25.199
Independent participants	38.000

0.5 Compact metrics summary

```

mean_high <- mean(paired$high_tempo_tte)
mean_low <- mean(paired$low_tempo_tte)
effect_mean <- mean(paired$diff_high_minus_low)
sd_diff <- sd(paired$diff_high_minus_low)
t_metrics <- t.test(paired$high_tempo_tte, paired$low_tempo_tte, paired = TRUE)
percent_improvement <- 100 * effect_mean / mean_low
completion_rate_registered <- 100 * participant_flow[, sum(completed_both_observed)] / nrow(participant_flow)

metrics_tab <- data.table(
  metric = c(
    "Mean treatment effect (high - low, seconds)",
    "Percentage improvement vs low tempo mean",
    "Standard deviation of the paired difference",
    "Paired t-test p-value",
    "Completion rate (completed both / registered)"
  ),
  value = c(
    sprintf("%.2f", effect_mean),
    sprintf("%.1f%%", percent_improvement),
    sprintf("%.2f", sd_diff),
    sprintf("%.4f", t_metrics$p.value),
    sprintf("%.1f%%", completion_rate_registered)
  )
)

kable(metrics_tab, caption = "Requested summary metrics")

```

Table 3: Requested summary metrics

metric	value
Mean treatment effect (high - low, seconds)	14.37
Percentage improvement vs low tempo mean	16.3%
Standard deviation of the paired difference	28.92
Paired t-test p-value	0.0041
Completion rate (completed both / registered)	69.0%

0.6 Exploratory figures

0.6.1 Spaghetti plot

```
plot_long <- melt(
  paired,
  id.vars = c("participant_id", "diff_high_minus_low"),
  measure.vars = c("low_tempo_tte", "high_tempo_tte"),
  variable.name = "condition",
  value.name = "seconds"
)

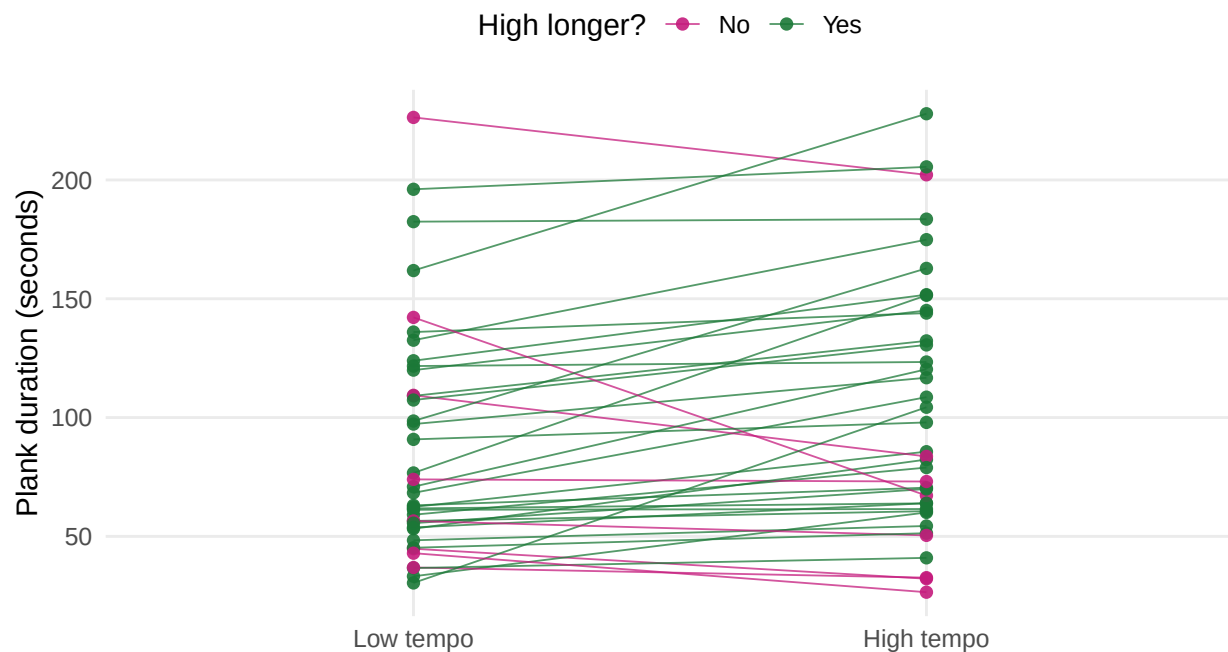
plot_long[, condition := factor(
  condition,
  levels = c("low_tempo_tte", "high_tempo_tte"),
  labels = c("Low tempo", "High tempo")
)]

plot_long[, favors_high := diff_high_minus_low > 0]

ggplot(plot_long, aes(x = condition, y = seconds, group = participant_id)) +
  geom_line(aes(color = favors_high), linewidth = 0.35, alpha = 0.75) +
  geom_point(aes(color = favors_high), size = 1.8, alpha = 0.9) +
  scale_color_manual(
    name = "High longer?",
    values = c(`TRUE` = "#1b7837", `FALSE` = "#c51b7d"),
    labels = c("FALSE" = "No", "TRUE" = "Yes")
  ) +
  labs(
    title = "Paired plank times by music condition",
    subtitle = paste0("n = ", uniqueN(plot_long$participant_id), " primary-sample participants"),
    x = NULL,
    y = "Plank duration (seconds)"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "top", panel.grid.minor = element_blank())
```

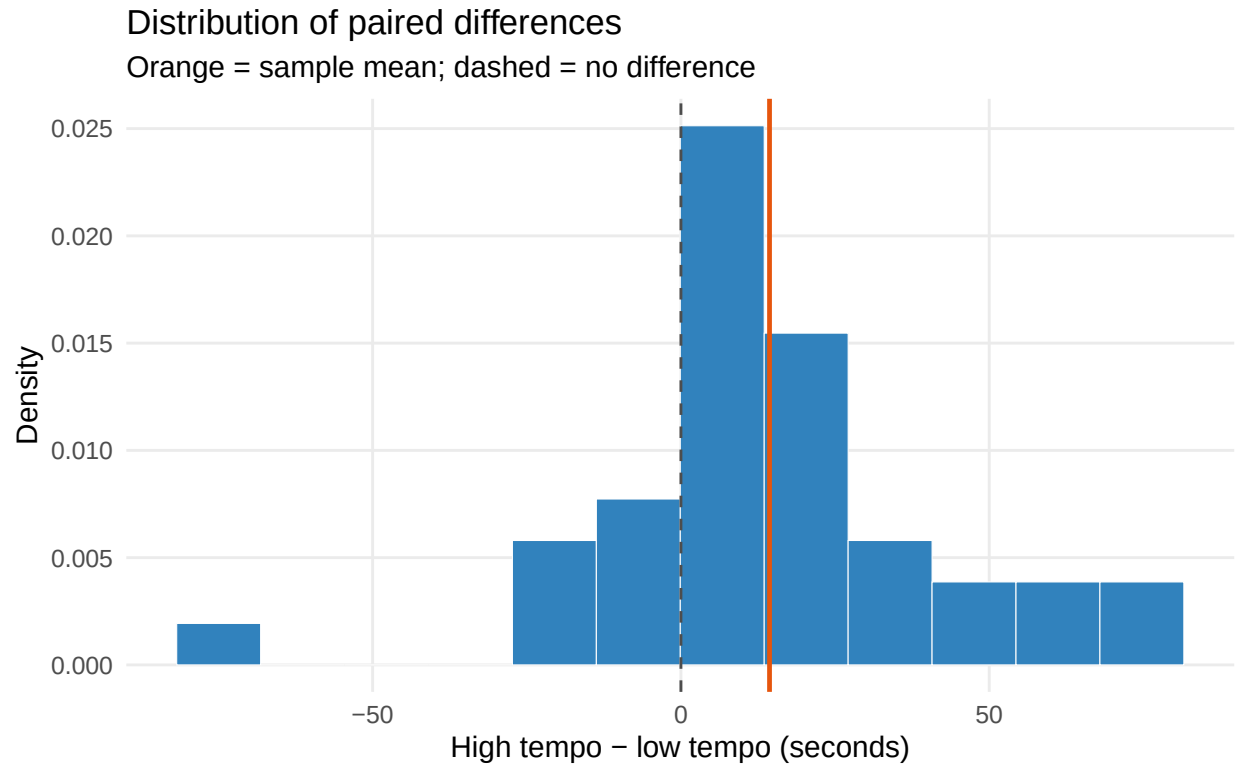
Paired plank times by music condition

n = 38 primary-sample participants



0.6.2 Paired-difference distribution

```
ggplot(paired, aes(x = diff_high_minus_low)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 12,
    fill = "#3182bd",
    color = "white",
    linewidth = 0.2
  ) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray30") +
  geom_vline(
    xintercept = mean(paired$diff_high_minus_low),
    color = "#e6550d",
    linewidth = 0.9
  ) +
  labs(
    title = "Distribution of paired differences",
    subtitle = "Orange = sample mean; dashed = no difference",
    x = "High tempo - low tempo (seconds)",
    y = "Density"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank())
```



0.7 Model 1: raw paired contrast

```
t_paired <- t.test(paired$high_tempo_tte, paired$low_tempo_tte, paired = TRUE)
effect_se <- sd(paired$diff_high_minus_low) / sqrt(nrow(paired))
effect_ci <- unname(t_paired$conf.int)
effect_dz <- effect_mean / sd(paired$diff_high_minus_low)

paired_results <- data.table(
  quantity = c(
    "Mean high-tempo TTE",
    "Mean low-tempo TTE",
    "Mean effect (high - low)",
    "95% CI lower",
    "95% CI upper",
    "Paired t statistic",
    "Paired p-value",
    "Standard error",
    "Paired effect size dz"
  ),
  value = c(
    mean(paired$high_tempo_tte),
    mean(paired$low_tempo_tte),
    effect_mean,
    effect_ci[1],
    effect_ci[2],
    unname(t_paired$statistic),
    t_paired$p.value,
    effect_se,
    effect_dz
  )
)
```

```

    effect_dz
  )
)

paired_results[, value := round(value, 4)]
kable(paired_results, caption = "Model 1. Raw paired contrast")

```

Table 4: Model 1. Raw paired contrast

quantity	value
Mean high-tempo TTE	102.4404
Mean low-tempo TTE	88.0709
Mean effect (high - low)	14.3694
95% CI lower	4.8633
95% CI upper	23.8756
Paired t statistic	3.0628
Paired p-value	0.0041
Standard error	4.6916
Paired effect size dz	0.4968

0.8 Model 2: fixed-effects crossover regression without period adjustment

```

long_primary[, participant_id := factor(participant_id)]

fe_noperiod_fit <- lm(
  outcome_tte ~ high_tempo + participant_id,
  data = long_primary
)

fe_noperiod_vcov <- sandwich::vcovCL(fe_noperiod_fit, cluster = ~ participant_id, type = "HC1")
fe_noperiod_ct <- lmtest::coeftest(fe_noperiod_fit, vcov. = fe_noperiod_vcov)

beta_high_noperiod <- unname(fe_noperiod_ct["high_tempo", "Estimate"])
se_high_noperiod <- unname(fe_noperiod_ct["high_tempo", "Std. Error"])
ci_high_noperiod <- beta_high_noperiod + c(-1, 1) * 1.96 * se_high_noperiod

fe_noperiod_results <- data.table(
  term = "High-tempo indicator",
  estimate = beta_high_noperiod,
  std_error = se_high_noperiod,
  conf_low = ci_high_noperiod[1],
  conf_high = ci_high_noperiod[2],
  p_value = unname(fe_noperiod_ct["high_tempo", "Pr(>|t|)"])
)

fe_noperiod_results[, `:=`(
  estimate = round(estimate, 4),
  std_error = round(std_error, 4),
  conf_low = round(conf_low, 4),
  conf_high = round(conf_high, 4),
  p_value = round(p_value, 4)
)]

```

```
kable(fe_noperiod_results, caption = "Model 2. Fixed-effects crossover regression without period adjustment")
```

Table 5: Model 2. Fixed-effects crossover regression without period adjustment

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.6796	1.2774	27.4615	0.0381

0.9 Model 3: fixed-effects crossover regression with period adjustment

```
long_primary[, period_factor := factor(period)]

fe_fit <- lm(
  outcome_tte ~ high_tempo + period_factor + participant_id,
  data = long_primary
)

fe_vcov <- sandwich::vcovCL(fe_fit, cluster = ~ participant_id, type = "HC1")
fe_ct <- lmtest::coeftest(fe_fit, vcov. = fe_vcov)

beta_high <- unname(fe_ct["high_tempo", "Estimate"])
se_high <- unname(fe_ct["high_tempo", "Std. Error"])
ci_high <- beta_high + c(-1, 1) * 1.96 * se_high

beta_period2 <- unname(fe_ct["period_factor2", "Estimate"])
se_period2 <- unname(fe_ct["period_factor2", "Std. Error"])
ci_period2 <- beta_period2 + c(-1, 1) * 1.96 * se_period2

fe_results <- data.table(
  term = c("High-tempo indicator", "Period 2 indicator"),
  estimate = c(beta_high, beta_period2),
  std_error = c(se_high, se_period2),
  conf_low = c(ci_high[1], ci_period2[1]),
  conf_high = c(ci_high[2], ci_period2[2]),
  p_value = c(
    unname(fe_ct["high_tempo", "Pr(>|t|)"]),
    unname(fe_ct["period_factor2", "Pr(>|t|)"])
  )
)

fe_results[, `:=`(
  estimate = round(estimate, 4),
  std_error = round(std_error, 4),
  conf_low = round(conf_low, 4),
  conf_high = round(conf_high, 4),
  p_value = round(p_value, 4)
)]

kable(fe_results, caption = "Model 3. Fixed-effects crossover regression with clustered SEs")
```

Table 6: Model 3. Fixed-effects crossover regression with clustered SEs

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.7101	1.2177	27.5212	0.0391
Period 2 indicator	3.8441	6.7101	-9.3076	16.9958	0.5703

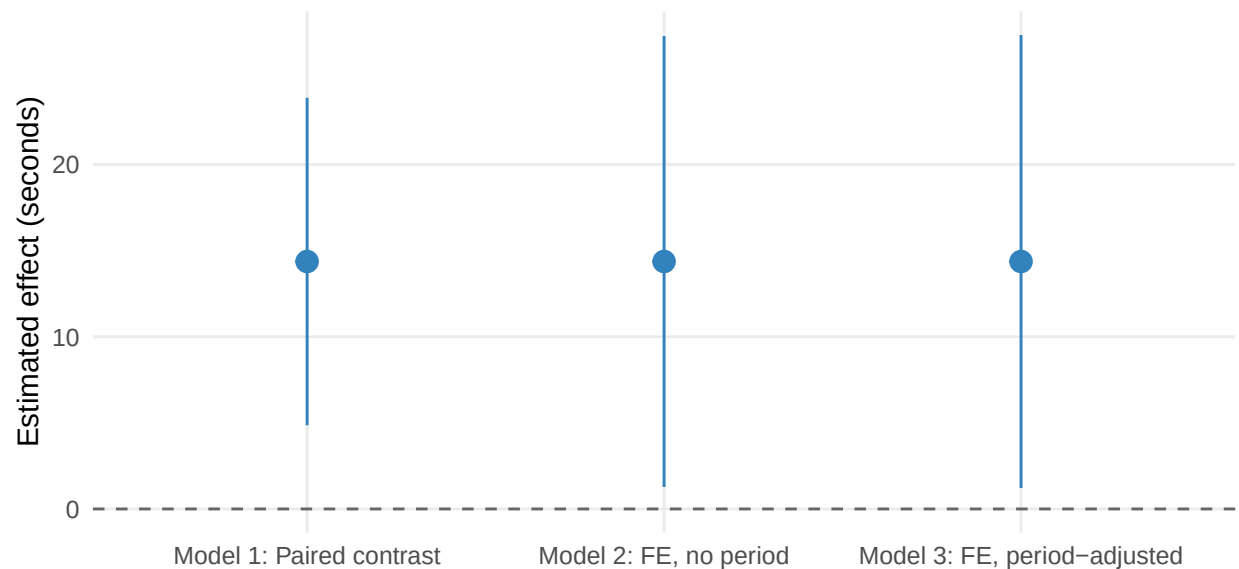
0.10 Main-effect comparison figure

```
coef_plot <- rbind(
  data.table(
    model = "Model 1: Paired contrast",
    estimate = effect_mean,
    conf_low = effect_ci[1],
    conf_high = effect_ci[2]
  ),
  data.table(
    model = "Model 2: FE, no period",
    estimate = beta_high_noperiod,
    conf_low = ci_high_noperiod[1],
    conf_high = ci_high_noperiod[2]
  ),
  data.table(
    model = "Model 3: FE, period-adjusted",
    estimate = beta_high,
    conf_low = ci_high[1],
    conf_high = ci_high[2]
  )
)
coef_plot[, model := factor(model, levels = model)]

ggplot(coef_plot, aes(x = model, y = estimate, ymin = conf_low, ymax = conf_high)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray40") +
  geom_pointrange(color = "#3182bd", linewidth = 0.55, size = 0.75) +
  labs(
    title = "Primary effect estimates",
    subtitle = "Effect on plank time in seconds (high tempo - low tempo) across the three-model ladder",
    x = NULL,
    y = "Estimated effect (seconds)"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank())
```

Primary effect estimates

Effect on plank time in seconds (high tempo – low tempo) across the three-model ladder



0.11 Verification

```
stopifnot(nrow(paired) == 38L)
stopifnot(uniqueN(long_primary$participant_id) == nrow(paired))
stopifnot(all(long_primary$in_primary_analysis == 1L))

message("Primary analytic participants: ", nrow(paired))
## Primary analytic participants: 38
message("Repeated-measures rows: ", nrow(long_primary))
## Repeated-measures rows: 76
message("Paired mean difference (high - low): ", round(effect_mean, 3))
## Paired mean difference (high - low): 14.369
message("FE without-period high-tempo coefficient: ", round(beta_high_noperiod, 3))
## FE without-period high-tempo coefficient: 14.369
message("FE high-tempo coefficient: ", round(beta_high, 3))
## FE high-tempo coefficient: 14.369
```